

Modelling generative AI's influence on students' perceived decision capability: A cognitive load and decision augmentation approach

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ABSTRACT

There is a resonant association between the implementation of Generative Artificial Intelligence (GenAI) and students' ability to identify, analyse, and select the best options among alternatives. Accordingly, this study examined the impact of GenAI adoption on business students' perception of their decision-making capability through the lenses of decision augmentation theory (DAT) and cognitive load theory (CLT). A survey was administered to 350 university students studying business in the United Arab Emirates. Using structural equation modelling, the study examined key constructs, including cognitive load reduction, information quality, and decision confidence as mediators, and GenAI trust as a moderator. The results revealed that GenAI significantly improves business students' perceived decision capability by improving confidence in decision-making and information quality. Although GenAI effectively reduces cognitive load, this reduction is negatively associated with students' perceived decision capability. Mediation analysis confirmed that decision confidence and information quality significantly mediate the relationship between GenAI adoption and perceived decision capability. Additionally, moderation analysis revealed that high trust in GenAI weakens the positive effects of GenAI on cognitive load reduction and information quality. These findings emphasise the importance of balanced GenAI integration, critical user engagement, and responsible trust development. The study provides valuable insights for students, educators, academic institutions, and industries seeking to use GenAI tools for enhanced learning and decision-making, and identifies limitations and directions for future research on GenAI-enabled education.

1. Introduction

Generative Artificial Intelligence (GenAI) is a branch of AI that involves producing new and original content, such as texts, images, videos, scenarios, and programming code, based on patterns learned from massive, diverse datasets (Feuerriegel et al., 2024). Compared with traditional AI systems, which were primarily designed for classification or prediction, contemporary GenAI models (e.g., Generative Pre-trained Transformer (GPT) for text or DALL-E for images) can generate human-like content (Kanbach et al., 2024; Lv, 2023). These models rely on deep learning algorithms, such as neural networks and transformers, that can mimic human creativity and language (Mariani & Dwivedi,

2024). Moreover, GenAI's ability to generate realistic, context-driven outputs makes it a powerful tool for enhancing productivity and creativity across domains (Chen et al., 2023). GenAI applications such as chatbots, ChatGPT, and virtual assistants are widely used in industry and academia (Holmström & Carroll, 2024).

The integration of GenAI in academia has significantly improved students' knowledge-building efficiency (Chen et al., 2023). Throughout their educational journey, university students deal with lengthy, complex, and information-intensive tasks. GenAI can amplify students' transformative potential, strengthening their cognitive abilities and intellectual capacities (Gkintoni et al., 2025). GenAI also helps students generate relevant, structured content that supports their academic and

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career advancement (Chen et al., 2023). Additionally, relying on GenAI reduces students' cognitive load (the amount of mental effort required by learners' working memory while processing information for their academic tasks; Jose et al., 2025; Sweller, 2011) by analysing and interpreting information and improving their perceived decision capability (Gutiérrez-Leefmans et al., 2025). That is, automating routine data processing and generating well-articulated presentations reduce students' mental effort, which in turn strengthens their decision confidence (belief in their own decisions) and improves their perceived decision capability (learners' self-belief in their decision-making competence; Albashrawi, 2025; Gkintoni et al., 2025; Koc-Januchta et al., 2022).

Previous studies have found that risks associated with GenAI use can undermine students' trust in AI-generated content, shape their learning attitudes, inhibit their cognitive abilities, diminish their perceived academic productivity, and potentially impact their academic outcomes (Arroyo-Bello et al., 2025; Samala et al., 2025). However, pedagogical studies have shown that the generation of well-organised, accurate, and context-driven information not only improves students' personalised learning (Lee et al., 2024) but also strengthens their decision-making ability in professional or entrepreneurial ventures (Duong & Nguyen, 2024). Students' trust and distrust in GenAI are critical factors in its adoption and use for future learning and decisions (Lyu et al., 2025). Literature has indicated that ethical concerns, privacy issues, and information accuracy associated with GenAI create doubts among its users (Chan & Hu, 2023; Gkintoni et al., 2025). However, students are optimistic about the use of GenAI in their academic endeavours since it contributes to improving their personalised learning (Chan & Lee, 2023), helps reduce their cognitive strain, and improves their confidence in using technology for education-related decision-making (Viberg et al., 2024).

Recent research across academic disciplines has examined the ethical aspects, benefits, and drawbacks of GenAI use (Chan & Hu, 2023; Wang et al., 2023). Alkaissi and McFarlane (2023) discussed how the illogical and irrelevant content sometimes generated by ChatGPT influences scientific academic writing, while Hannigan et al. (2024) found that the content that GenAI chatbots provided is often irrelevant and vague. Other studies have focused on the benefits of using GenAI in academia, such as improvements in learning, problem-solving, and cognitive competency (Gupta & Jaiswal, 2024; Jose et al., 2024). Beyond the existing literature, there remains a gap in examining the mechanisms by which GenAI-supported environments affect students' perceived decision capability through cognitive factors such as cognitive load reduction and decision confidence. Although information quality is widely recognized in information systems research as a critical factor in effective decision-making, the role of AI-generated information quality in academic decision-making remains underexplored. In addition, few studies have integrated Decision Augmentation theory (DAT) and Cognitive Load theory (CLT) in explorations of how GenAI-based tools simultaneously function as decision support systems and cognitive support mechanisms in educational settings.

According to Jose et al. (2025), integrating GenAI into the learning process is both a form of technological integration and a cognitive and psychological transformation that influences students' perceived ability to make decisions (Ododo et al., 2024). Understanding these interconnections is essential for educators and institutions aiming to leverage GenAI as a decision-enhancing educational tool (Patac & Patac, 2025). Therefore, this study aims to investigate how business students' trust in GenAI leads to an improved cognitive experience, higher-quality information, and greater confidence in their decisions. The use of CLT and DAT in education is grounded in the understanding that GenAI does not replace critical thinking but augments students' analytical ability, thereby enhancing the perceived quality of their judgments (Jose et al., 2025).

To address these knowledge gaps, this study aims to answer the following research questions:

RQ1. *How is GenAI adoption associated with business students' cognitive load reduction, information quality, decision confidence, and perceived decision capability?*

RQ2. *How are business students' cognitive load reduction, information quality, and decision confidence associated with their perceived decision capability?*

RQ3. *Do business students' cognitive load reduction, information quality, and decision confidence mediate the association between GenAI adoption and perceived decision capability?*

RQ. *Does business students' GenAI trust moderate the association between GenAI adoption and cognitive load reduction, information quality, and decision confidence?*

The aim of this research is to study students' perception of decision-making capability and confidence in academic decisions when using GenAI tools, rather than measuring their objective decision accuracy. In the context of GenAI-empowered decision support systems, perceived decision capability is a crucial behavioural outcome since users' trust, confidence, and technology usage largely depend on their subjective evaluations of system outputs (El-Amran et al., 2025; McKnight et al., 2011). Hence, the study has three key objectives: (a) to develop a conceptual model that connects various variables in the student learning process based on DAT and CLT; (b) to substantiate how load reduction, information quality, and decision confidence as mediators improve business students' perceived decision capability with the adoption of GenAI; (c) to verify whether GenAI trust as a moderating variable has any influence on the relationship between business students' GenAI adoption and cognitive load reduction, information quality, and decision confidence.

2. Theoretical framework

2.1. Decision augmentation theory

Introduced by May et al. (1995), DAT explains how individuals' decision-making ability can be augmented through automated systems that enhance cognitive processes (May et al., 1995; Tao et al., 2024). DAT emphasises the role of decision support systems (e.g., AI and data analytics) in augmenting individuals' cognitive abilities to achieve better decision-making outcomes (Leyer & Schneider, 2021). The adoption of DAT in academia helps in understanding students' competencies by leveraging GenAI in academic learning environments (Gupta, Mahajan, et al., 2024; Harfouche et al., 2023). Associated with DAT's core principle of symbiotic decision-making, students learn to integrate human cognition with machine intelligence through classroom applications, case studies, and real-time simulations (Herath Pathirannehelage et al., 2025). The adoption of GenAI expands students' abilities, such as analysis of large datasets, scenario modelling, and assessment of results with greater precision (Suryanarayana et al., 2024). It also improves their skills in making strategic decisions based on data analytics and in identifying future opportunities and obstacles. Additionally, it streamlines bulk information, reducing students' cognitive load and allowing them to focus on higher-priority tasks (Burton et al., 2020). Decision augmentation tools can also improve students' confidence when dealing with complex or delicate situations encountered during their academic journey (Herath Pathirannehelage et al., 2025).

2.2. Cognitive load theory

CLT posits that humans' short-term memory (i.e., working memory) has a limited capacity and can handle only a small amount of information at a time (Sweller, 1988, 2011). Contrastingly, long-term memory can store large amounts of information in structured forms known as schemas (De Jong, 2010). Learning can be effective when information stored in working memory is successfully transferred to long-term memory through schema development (Paas et al., 2020). According

to CLT, three types of cognitive load exist: (a) intrinsic load, which refers to the inherent complexity of the information; (b) extraneous load, which arises from poor instructional design that distracts or confuses learners; and (c) germane load, which denotes the cognitive effort directed toward understanding and organising material to form schemas (Patac & Patac, 2025). CLT is key to understanding how advanced technology (i.e., GenAI) can be used to optimise students' learning by aligning with their cognitive schemas (Jose et al., 2025). In the context of business students' adoption of GenAI, this interpretation suggests that advanced learning tools reduce extraneous load, help in managing intrinsic load (i.e., task alignment), and enhance germane load to promote deep learning while supporting strategic decision-making (Gkintoni et al., 2025; Patac & Patac, 2025).

3. Literature review and hypotheses development

3.1. GenAI in management education

In a dynamic academic environment, students' use of advanced technological tools to help them achieve their educational outcomes enhances their critical thinking, which, in turn, strengthens their trust in these tools (Gupta, Mahajan, et al., 2024; Huy et al., 2024). Students' adoption of GenAI is significantly associated with their perceived knowledge and the perceived threat of the technology (Al-Emran et al., 2025; Herath Pathirannehelage et al., 2025). Sze et al. (2026) found a direct relationship between the benefits of GenAI and students' adoption intentions. However, the risks associated with GenAI do not directly influence students' behavioural intentions. Changalima et al. (2024) found that students use GenAI to obtain reliable information, data-driven insights, and real-time feedback, supporting iterative learning that builds their confidence in academic and professional decisions. Such refined results based on quality information help students develop effective strategies for their educational endeavours. Xia et al. (2024), on the other hand, demonstrated that students use GenAI to analyse complex problems, generate potential solutions, and evaluate outcomes quickly, thereby enhancing their academic abilities and improving their learning efficiency. Students' adoption of GenAI can also foster collaborative learning by offering diverse perspectives and synthesising information from multiple sources (Wood & Moss, 2024). Moreover, Vieriu and Petrea (2025) argued that the quality of information produced by GenAI enhances students' critical thinking, improving their decision-making ability in a rapidly changing academic and business world. Therefore, students' use of GenAI not only helps them manage their cognitive load but also improves their learning efficiency, ultimately augmenting their perceived decision capability (Herath Pathirannehelage et al., 2025; Patac & Patac, 2025).

3.2. GenAI adoption and cognitive load reduction

Cognitive load reduction refers to reducing students' mental efforts when attempting to comprehend information in working memory, thereby enhancing learning efficiency and decision-making capability (De Jong, 2010). The adoption of GenAI among students can transform their overall learning experience by significantly reducing the cognitive and workloads associated with completing tasks (Patac & Patac, 2025). GenAI reduces students' cognitive load by assisting them with prompt and efficient task completion (i.e., analysing data, writing reports, and preparing presentations). It can also minimise their workload by automating routines and completing repetitive tasks such as summarising lengthy articles or reflecting on case studies. In both respects, GenAI encourages students to apply their analytical reasoning, which enhances their perceived decision capability (Bahroun et al., 2023). On the other hand, reducing the number of tasks to perform or the cognitive load required to process information not only enhances students' performance but also reduces their educational stress. Students are then better able to manage their time, allowing them to achieve desired outcomes

across academic, personal, and professional domains (Jose et al., 2024). However, Almugren, Rai, Korayim, Mohanty, and Virmani (2026) found that excessive use of GenAI in education hinders students' cognitive growth.

Gkintoni et al. (2025) integrated AI and Machine Learning with CLT, noting that GenAI can reduce students' extraneous load and increase their germane load by providing personalised learning experiences (e.g. through instant feedback, concept clarity, and simulated real-world scenarios). This can significantly improve students' learning efficiency. The study also depicted that, despite the use of advanced technology, ethical concerns, data security, and privacy issues remain relevant when using GenAI in academia. Similarly, Patac and Patac (2025) stated that students' adoption of GenAI (i.e., ChatGPT) for educational purposes significantly reduces their intrinsic and extrinsic cognitive load. The quality of information and the detailed immediate responses generated by GenAI prompts can reduce students' academic burden, enhance their understanding, and develop their critical thinking skills. Likewise, Jose et al. (2025) reported that the use of GenAI in education significantly reduces students' cognitive load, provides personalised guidance, and enhances learning, ultimately improving decision-making outcomes. However, they also reported that over-reliance on GenAI can undermine students' critical thinking ability, thereby affecting their overall academic performance. To avoid this drawback, they suggested that properly integrating GenAI tools in educational settings from course design can improve students' self-learning, foster their intellectual abilities, and amplify their cognitive thinking. Therefore, the following two hypotheses were formulated related to business students:

H1. Business students' adoption of GenAI is positively associated with their cognitive load reduction.

H1a. Cognitive load reduction mediates the relationship between business students' adoption of GenAI and their perceived decision capability.

3.3. GenAI adoption and information quality

In the context of management education and the scope of this study, information quality refers to information being accurate, relevant, timely, and reliable in fulfilling the specific requirements and intended use of business students (Changalima et al., 2024). In academia, GenAI tools (e.g., ChatGPT, Gemini, Copilot, Perplexity, DeepSeek) are increasingly shaping students' educational performance through interactive, personalised learning that directly influences their motivation (Jose et al., 2024). A pivotal reason for this dependency is the perceived quality of information provided by the AI (Changalima et al., 2024). However, a lack of proper guidance on the use of advanced technological tools creates challenges for students in verifying AI outputs, including inappropriate or vague information, information bias, and data privacy concerns (Alkaissi & McFarlane, 2023; Hannigan et al., 2024). Only when students become AI-literate and can write effective prompts to search for information can they critically evaluate the outputs of GenAI to determine whether they are credible and contextually appropriate (Chan & Hu, 2023). They can then focus on the practical application of this information in preparing for their academic tasks, such as writing reports, performing analyses, and completing exercises (Pallant et al., 2025). Information quality also enhances students' cognitive efficiency by reducing the need to cross-verify facts or reinterpret vague content, a result consistent with the core ideas of CLT (Jose et al., 2025).

Changalima et al.'s (2024) findings indicated that meticulous information quality strengthens students' behavioural intentions to adopt and use GenAI tools in academic contexts. Drawing on DAT, once students' cognitive knowledge is complemented by high-quality AI-based information, their learning efficiency and innovative inspiration are enhanced (Chen et al., 2023; Herath Pathirannehelage et al., 2025). Almugren, Rai, Chotia, Szabó-Szentgróti, and Mohanty (2026) found

that GenAI provides students with quality information, which is positively related to their entrepreneurial passion and readiness. Kuo-Wei (2025) noted the transformative role of GenAI-based information sources in enhancing business students' learning, helping them acquire diverse academic and practical knowledge beneficial for future decision-making. Previous studies have also demonstrated that high-quality AI-generated information enhances students' knowledge and capacity to interpret and respond rapidly to a dynamic educational environment (Chen et al., 2023; Jose et al., 2024). In light of the aforementioned review, the study develops the following two hypotheses about business students:

H2. Business students' adoption of GenAI is positively associated with GenAI information quality.

H2a. GenAI information quality mediates the relationship between business students' adoption of GenAI and their perceived decision capability.

3.4. GenAI adoption and decision confidence

Koriat et al. (1980) defined decision confidence as the degree to which a person believes that his or her answer or choice is correct. Decision confidence is a metacognitive judgement that plays a central role in decision-making processes, influencing subsequent behaviours such as willingness to act, additional information seeking, or previous decision revision (Phillips et al., 2014). Based on CLT, Gkintoni et al. (2025) reported that AI-based instructional methods reduce students' cognitive load and improve their learning efficiency through personalised and adaptive learning, thereby enhancing their confidence in making academic decisions (Patac & Patac, 2025). Zhang et al. (2026) found that students' use of GenAI for educational purposes is positively associated with knowledge acquisition and application, which they use to develop their entrepreneurial skills and intentions. However, risks associated with the use of GenAI tools, such as data privacy issues, cybersecurity threats, misinformation and manipulation, bias, and lack of transparency, create distrust in and doubt about the credibility of the information produced (Chan & Hu, 2023; Sze et al., 2026), thereby impacting users' decision confidence.

Chan and Lee (2023) noted that, in critical circumstances, GenAI offers substantial value by providing real-time data synthesis, scenario modelling, and strategic insights that help students make important decisions. In a similar study, Hao et al. (2024) found that GenAI delivers reliable, relevant, and context-driven information to users, thereby improving their decision confidence. This confidence acts as a cognitive facilitator, enabling users to approach case-based problems and simulations with greater clarity, reduced ambiguity, and improved analytical depth (Gkintoni et al., 2025). In another study, Zhang et al. (2020) observed that decisions based on AI models outputs are not sufficient until users are prepared to manage AI risks. Drawing from the preceding findings from published articles, the study proposes the following two hypotheses concerning business students:

H3. Business students' adoption of GenAI is positively associated with their decision confidence.

H3a. Decision confidence mediates the relationship between business students' adoption of GenAI and their perceived decision capability.

3.5. GenAI adoption and perceived decision capability

GenAI aids in developing decision-makers' analytical, strategic, and creative thinking abilities (Herath Pathirannehelage et al., 2025; Pallant et al., 2025). Several scholars have shown that GenAI can enhance users' perceived decision-making capability by providing relevant and structured content, which improves their confidence and effectiveness in making complex decisions (Albashrawi, 2025; Hao et al., 2024). Yilmaz and Yilmaz (2023) also found that GenAI can improve students'

computational skills and enthusiasm. Furthermore, GenAI enables students to process and analyse large datasets efficiently, leading to more informed, evidence-based decisions. With development tools such as predictive modelling and scenario simulations, students can generate a range of academic business documents (e.g., business plans, research proposals, customer interviews), identify surrounding risks (i.e., analysing internal and external factors), and use these insights to strengthen their strategic planning skills (Vecchiarini & Somià, 2023). However, in addition to the potential benefits of GenAI, studies have identified threats to knowledge acquisition when information sources are unreliable, misleading, or distorted (Chan & Hu, 2023) since this can adversely influence students' decision-making efficiency (Zhang et al., 2020).

In accordance with CLT and DAT, Patac and Patac (2025) and Herath Pathirannehelage et al. (2025) posited that GenAI tools can reduce the cognitive load required to process vast and complex information, ultimately improving students' learning and perceived decision capability. Jose et al. (2024) found that when students have positive perceptions of GenAI, they believe that GenAI can equip them with high-quality information, motivating them to use it for management-based educational decision-making. Yang et al. (2025) found that the use of GenAI in medicine directly influences medical practitioners' perceived decision-making ability. In another study, Alghamdi (2025) found that students' positive perception of the use of GenAI in academia enhances their critical thinking and improves their creative abilities. Similarly, Changalima et al. (2024) found that GenAI can enhance information quality by generating data-driven insights, summaries, and analyses, helping students access relevant, accurate, and timely information, which, in turn, boosts their perceived decision capability (Alghamdi, 2025). Furthermore, Chen et al. (2023) argued that GenAI can provide students with reliable support to enhance their knowledge-building capacity, helping them make the right decision in a timely manner, leading to better outcomes. Following this review, the study identifies the following hypothesis regarding business students:

H4. Business students' adoption of GenAI is positively associated with their perceived decision capability.

3.6. Cognitive load reduction and perceived decision capability

In academia, load reduction refers to easing students' mental load through the efficient use of tools or strategies, thereby improving comprehension and performance (Sweller & Chandler, 1991; van Merriënboer & Sweller, 2005). Several studies have shown that refined instructional strategies help reduce cognitive load among students and improve their engagement and learning outcomes. Reductions in mental effort enhance individuals' perceived ability to perform tasks effectively (Evans et al., 2024). In the evolving landscape of management education, GenAI is playing a pivotal role in transforming business students' academic approaches and decision-making processes. Patac and Patac (2025) found that students' use of GenAI (i.e., ChatGPT) significantly reduces intrinsic and extraneous load by streamlining time-consuming tasks (i.e., literature reviews, data interpretation, content generation, and problem analysis), thereby enhancing learning efficiency and improving decision-making skills. However, overreliance on GenAI reduces students' critical thinking ability and memory retention (Bai et al., 2023). Chen et al. (2023) stated that by automating routine cognitive functions (i.e., problem identification, idea generation, evaluation of ideas, feasibility of ideas, and higher-level ideas) and providing real-time structured information, GenAI helps students focus more on their higher-order thinking skills, such as strategic evaluation, critical analysis, and ethical reasoning (overreliance on AI and information bias). In line with CLT and DAT, GenAI, as an amplifier, enhances students' knowledge-building capacity, reduces their extraneous load, and allows them to apportion mental resources toward deep learning, thereby augmenting their decision-making skills (Gkintoni

et al., 2025). Based on CLT, Jose et al. (2024) stated that incorporating GenAI tools into education can allow students to acquire advanced knowledge that enhances their cognitive capacity and motivates them to adopt the technology to achieve academic outcomes (Evans et al., 2024). Grounded in the insights gathered above, the research posits the following hypothesis related to business students:

H5. Business students' cognitive load reduction is positively associated with their perceived decision capability.

3.7. Information quality and perceived decision capability

Information quality in management education plays a crucial role in shaping students' decision-making capability, widely regarded as a core competency for future business leaders (Herath Pathirannehelage et al., 2025). In today's academic environment, information quality refers to how effectively students can process and apply data to understanding real-world problems (Pallant et al., 2025). For students engaged in tasks such as content planning, market analysis, forecasting, and critical discourse (Chen et al., 2023), the ability to access high-quality information determines not only the reliability of their judgments but also their confidence and learning efficiency in decision-making (Vieriu & Petrea, 2025). Despite these benefits, studies have found that vague, contextually irrelevant, misleading, and biased information (Alkaissi & McFarlane, 2023; Hannigan et al., 2024), often generated by GenAI tools, can inhibit students' learning and problem-solving abilities (Çela et al., 2024).

Tao et al. (2024) suggested that high-quality inputs lead to more optimal outputs; thus, the credibility of the information used directly impacts the quality of decisions. In a digital learning environment supported by GenAI, the perceived quality of AI-generated information augments students' decision-making skills (Leyer & Schneider, 2021). Almugren, Rai, Korayim, et al. (2026) found that GenAI-based information literacy negatively influences students' level of motivation and has no impact on their behavioural intentions. Pallant et al. (2025) argued that educators should emphasise not only the acquisition of information but also its quality. They further demonstrated this by teaching students to assess GenAI tools critically, triangulate data, and interpret information in context, thereby augmenting their analytical acumen and preparing them to develop effective decision-making skills. Given the review of the prior three studies, the study arrives at the following hypothesis involving business students:

H6. GenAI information quality is positively associated with business students' perceived decision capability.

3.8. Decision confidence and perceived decision capability

Decision confidence refers to individuals' trust in the accuracy and relevance of their judgments (Arts et al., 2024). A confident decision-maker is more capable of evaluating alternatives, executing solutions assertively, and being accountable for outcomes (Moore & Healy, 2008). In academic contexts, decision confidence plays a crucial role in enhancing individuals' decision-making ability (Grimaldi et al., 2015); however, few studies have examined the direct association between students' decision confidence and perceived decision capability. Moore and Healy (2008) found that greater confidence in decision-making improves individuals' perceptions of behavioural capability since people who feel more certain about their decisions tend to believe more strongly in their ability to perform tasks effectively. In a multidisciplinary context, Albashrawi (2025) reported that GenAI contributes positively to individuals' decision-making. Contrastingly, Oblakovic et al. (2025) found that GenAI strengthens students' meta-cognitive confidence while reducing their decision-making ability. Yeung and Summerfield (2012) suggested that decision confidence enables individuals to adjust their decision strategies in real time, leading to more efficient and adaptive decision-making. They further showed

that decision confidence improves performance in high-stakes and uncertain environments by reducing hesitation, streamlining cognitive processing, and supporting the better calibration of judgment (Ishizu et al., 2024). Bromberg-Martin and Sharot (2020) stated that decision confidence not only improves the quality of immediate decisions but also fosters students' long-term self-regulated learning and reflective thinking, both of which are critical for higher education students (McDonald et al., 2025). Moreover, Pallant et al. (2025) found that students' learning is better when they use GenAI to augment their knowledge, as reflected in their academic decisions and outcomes. From the preceding review, the study decided to test the following hypothesis related to business students:

H7. Business students' decision confidence is positively associated with their perceived decision capability.

3.9. GenAI trust as a moderator

Trust is "a willingness to rely on an exchange partner in whom one has confidence" (Moorman et al., 1992, p. 83). According to the International Organization for Standardization (2020), AI is considered a trustworthy source if it meets all the standard requirements of its users, as Kaur et al. (2022) posited: A trustworthy AI tool should provide fair, explainable, accountable, secure, and reliable information. Lyu et al. (2025) examined the degree of trust or distrust and psychological assurance regarding next-generation technologies (AI) that help or hinder their users (students or professionals) in performing particular tasks accurately, reliably, ethically, and as per their expectations and goals. Shin (2021) emphasised the importance of the explanatory power of AI and found that causability (what and how things should be explained), as an antecedent of AI explainability, builds users' trust in AI algorithms, which in turn affects their decision-making. From the perspective of CLT, Paas et al. (2020) stated that learners face constraints in their working memory capacity, which can be alleviated through adequate instructional support. When students adopt GenAI tools with high trust, they are more likely to perceive these technologies as reliable cognitive partners, thereby reducing their mental effort and streamlining their information processing (Patac & Patac, 2025). Reflecting on management education, trust in GenAI not only encourages its adoption but also determines whether its use meaningfully reduces students' cognitive load (Jose et al., 2025). Taking the earlier discussion into account, the following hypothesis on the moderation effect of GenAI trust is what this study aims to explore among business students:

H8a. GenAI trust moderates the relationship between business students' GenAI adoption and their cognitive load reduction.

Information quality characteristics, such as accuracy, relevance, timeliness, and completeness, predict technology adoption (Feuerriegel et al., 2024). Changelima et al. (2024) stated that students' positive perceptions of AI-generated information quality convince them to use these tools to support academic decision-making, learning efficiency, and cognitive engagement. However, the strength of this relationship is significantly influenced by the level of trust students place in GenAI (Amoozadeh et al., 2024). Previous studies have raised concerns about hallucinations (i.e., misleading, vague, and biased outputs) generated by GenAI (Alkaissi & McFarlane, 2023; Hannigan et al., 2024). Kaur et al. (2022) stated that to increase trust in GenAI tools, users must understand their applicability and limitations. Huynh and Aichner (2025) examined informational fairness (unbiased and non-discriminatory) as directly associated with trust in GenAI tools, which positively influences users' attitudes and behavioural intentions. Amoozadeh et al. (2024) argued that students with high trust in GenAI are more likely to interpret high-quality information as credible and actionable, thereby enhancing their willingness to adopt the technology. Contrastingly, students with low trust in GenAI may remain sceptical about the utility or accuracy of

such information, which in turn diminishes their intention to adopt GenAI (Lyu et al., 2025). This moderating effect aligns with the extended technology acceptance model, according to which trust acts as a psychological enabler that reduces perceived risk and uncertainty (Venkatesh et al., 2012). Based on the above arguments, the study proposes the following moderation effect of GenAI trust, as outlined in the following hypothesis:

H8b. GenAI trust moderates the relationship between business students' GenAI adoption and GenAI information quality.

Lukya-nenko et al. (2022) considered trust to be a critical aspect in GenAI adoption and usage for users. Lyu et al. (2025) stated that trust in advanced technologies serves as a key factor that influences students' relationship with GenAI and impacts their decision confidence. Zhang et al. (2020) defined decision confidence as the degree of certainty individuals feel about the correctness and reliability of their judgments. The study also found that confidence scores enhance users' trust in GenAI tools, but trust alone is not sufficient to improve decision-making unless users learn how to overcome the limitations of AI. Chen et al. (2023) stated that students' use of GenAI for academic decision-making (i.e., solving problems, generating insights, or evaluating alternatives) not only demonstrates their confidence but also their trust in GenAI (Huynh & Aichner, 2025). Amoozadeh et al. (2024) argued that when students exhibit a high level of trust in GenAI, the positive impact of adoption on decision confidence is significantly amplified. The study further disclosed that trust reduces students' perceived uncertainty and fosters a strong belief in GenAI being able to improve their performance expectancy (Huynh & Aichner, 2025). In the context of management education, where students must make complex, data-informed decisions, cultivating trust in GenAI ensures that the benefits of adoption translate effectively into elevated decision confidence, ultimately enhancing students' decisional capability and professional preparedness (Weng et al., 2024). Based on the above prior findings, the study proposed a moderation effect by GenAI trust, as designed in the following hypothesis:

H8c. GenAI trust moderates the relationship between business students' GenAI adoption and their decision confidence.

The conceptual framework of this study is presented in Fig. 1.

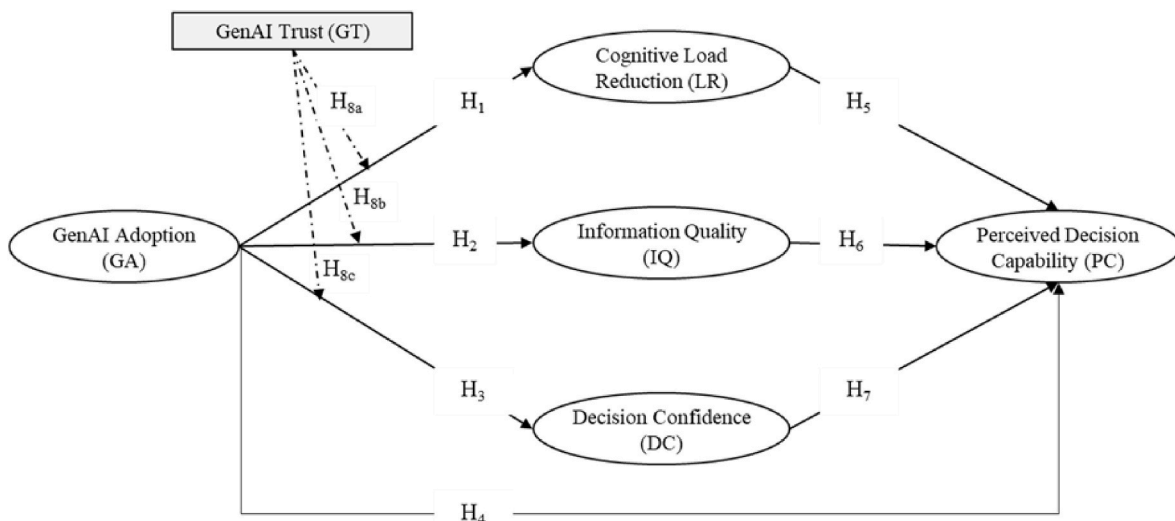


Fig. 1. Conceptual framework.

4. Research methodology

4.1. Measures

The study adopted standardized scales from existing research to collect data. The GenAI adoption scale applied was based on the scale from Singh et al.'s (2024) study. Scale items for load reduction were drawn from Martin et al.'s (2023) study, information quality from Chagalima et al.'s (2024) study, and decision confidence from Phillips et al.'s (2014) study. The perceived decision capability scale was adapted from that of Sadeghi et al. (2024), and the GenAI trust scale was obtained from Huy et al.'s (2024) study. The final questionnaire administered to students comprised 31 items, with responses given on a 7-point Likert scale ranging from 1 (strongly disagree) to 7 (strongly agree). These items were intended to examine the association between students' use of GenAI and the cognitive mechanisms (load reduction, information quality, and decision confidence) that ultimately influence their perceived decision capability. The latter was defined within a task-agnostic model rather than within a comparison of performance on a single business analytical task (e.g., a case study or project evaluation). Therefore, the current study aligns with previous research that operationalised decision-making constructs as perceived capability enhancements facilitated by GenAI tools rather than objective performance outcomes (i.e., grade assessment; Almogren et al., 2024).

4.2. Data collection

Data were collected from four recognized universities in the United Arab Emirates. Students included in the study were registered business students at the undergraduate or postgraduate level. A list of students' email addresses was obtained from the registrar's office of these universities (Ruihley et al., 2020; Rai et al., 2023; Rai et al., 2024). The questionnaire link was emailed to around 2300 students in total (Billings et al., 2016; Rai et al., 2021). Within 4 weeks, during which an email reminder to participate was sent, 412 responses were received. After a careful examination of responses and omission of incomplete and unengaged responses, 386 responses were retained for the next stages of analysis. Gender distribution was even for female and male respondents. Students' ages ranged from 18 to 24 years, and 64% were undergraduates. Most students used GenAI tools for educational purposes, mainly for business analytical tasks, though at varying percentages: 57% used them regularly, 32% sometimes, and 11% sporadically.

4.3. Data analysis procedure

A cut-off value of $p < 0.01$ was set for statistical significance across all parameters (Trafimow & Earp, 2017). Data normality was assessed by checking the skewness and kurtosis values for all variables. All values fell within the prescribed limits of -2 to $+2$ (for skewness) and -7 to $+7$ (for kurtosis), thereby indicating data normality (Curran et al., 1996). The presence of common method bias (CMB) in the data was assessed using two tests. First, Harman's Single Factor Test (Podsakoff et al., 2003) was used to assess the variance explained by a single factor. The computed single factor explained only 34.64% (less than 50%) of the variance in the data, suggesting a lack of CMB. Second, the common latent factor approach was used. The common method approach yielded $CMIN/df = 2.51$, whereas the measurement model yielded $CMIN/df = 2.697$. The difference in $CMIN/df$ was less than 0.2, suggesting a lack of CMB. Additionally, the difference in factor loadings for almost all observed variables was less than 0.20, further suggesting a lack of CMB.

Structural equation modelling (SEM) was used to analyse the relationships among multiple variables identified in the study. Since this study was deductive, aiming to test the validity of existing theories, the covariance-based (CB) technique was used (Hair et al., 2014). CB-SEM was appropriate for the confirmatory factor analysis (CFA) chosen for three main reasons. First, it provided a precise assessment of discriminant and convergent validity. Second, the constructs used in the study were adopted from previously developed scales. Third, it provided thorough goodness-of-fit indices that were unattainable using other statistical methods like partial least squares-SEM. Therefore, by presenting more verifiable features, CB-SEM lessened model estimation bias (Kline, 2016).

5. Results

The measurement and structural models are analysed using AMOS version 26. The moderation and mediation analyses were performed using the Process Macro Model 4.

5.1. Measurement model

CFA was applied to assess the reliability and validity of the first-order constructs identified in the study. CFA is a constrained model technique in which all cross-loadings are abandoned to confirm the relationships between constructs and their measurements. The Maximum Likelihood Estimation (MLE) technique was used to generate model parameters. The goodness-of-fit statistics were as follows: $CMIN = 1119.368$, $df = 415$, $CMIN/df = 2.697$, $CFI = 0.94$, $TLI = 0.933$, $IFI = 0.94$, $NFI = 0.91$, and $RMSEA = 0.066$ (<0.08), all of which indicated acceptable model fit.

As shown in Table 1, the factor loadings for all constructs exceeded 0.70, indicating that more than 50% of the variance in the observed variables was explained by their respective constructs (Hair et al., 2019) and supported the model's convergent validity. Furthermore, the AVEs for the constructs ranged from 0.692 to 0.796, well above the recommended minimum AVE of 0.50, thereby confirming the convergent validity of the constructs (Hair et al., 2019). The reliability of the constructs was assessed using Cronbach's alpha (CA) and composite reliability (CR). The CA statistics for the constructs ranged from 0.910 to 0.958, while the CR varied between 0.912 and 0.957. These reliability statistics exceeded the minimum threshold of 0.70 (Hair et al., 2019), providing strong evidence for high reliability.

Discriminant validity was assessed using the Fornell–Larcker criterion (Fornell & Larcker, 1981) to substantiate that each construct in the model was distinct and captured a phenomenon not measured by other constructs (Hair et al., 2019). The necessary condition for discriminant validity was that the square root of each construct's AVE exceeded the respective inter-construct correlation (Fornell & Larcker, 1981). As shown in Table 2, where diagonal values represent the square roots of

Table 1

Summary of measurement model results.

Construct	Measurement Item	Factor Loading	Mean (SD)
GenAI Adoption (GA) C-Alpha = 0.925 AVE = 0.711 CR = 0.925 MSV = 0.585	- GenAI adoption would lead to a substantial improvement in my academic productivity.	0.855	5.15 (1.53)
	- The utilisation of GenAI would enable me to accomplish complex tasks more efficiently.	0.829	5.33 (1.55)
	- I would find it easy to instruct GenAI to perform specific tasks.	0.814	5.27 (1.51)
	- GenAI would make it easier for me to perform my academic responsibilities.	0.868	5.23 (1.47)
	- Becoming proficient at using GenAI would be a straightforward and manageable endeavour.	0.849	5.10 (1.45)
Cognitive Load Reduction (LR) C-Alpha = 0.930 AVE = 0.720 CR = 0.928 MSV = 0.442	- GenAI helps to learn complex data in a simplified way.	0.859	5.16 (1.66)
	- GenAI provides help when I need it.	0.899	5.38 (1.68)
	- GenAI helps me to practice important tasks before actual implementation.	0.884	5.12 (1.54)
	- GenAI provides me constructive feedback.	0.840	4.99 (1.62)
	- GenAI helps me to make decisions independently.	0.755	4.81 (1.66)
Information Quality (IQ) C-Alpha = 0.958 AVE = 0.692 CR = 0.957 MSV = 0.609	- Information extracted through GenAI tools is believable.	0.824	4.88 (1.55)
	- Information extracted through GenAI tools is unbiased.	0.747	4.85 (1.55)
	- Information extracted through GenAI tools is of right amount.	0.832	4.90 (1.51)
	- Information extracted through GenAI tools is value-added.	0.858	5.08 (1.49)
	- Information extracted through GenAI tools is easy to interpret.	0.825	5.18 (1.50)
	- Information extracted through GenAI tools is well organised.	0.835	5.22 (1.54)
	- Information extracted through GenAI tools is consistent and sensible.	0.848	5.13 (1.45)
	- Information extracted through GenAI tools is in a concise format.	0.850	5.11 (1.49)
	- Information extracted through GenAI tools is easily available.	0.859	5.28 (1.50)
	- Information extracted through GenAI tools is easy to find.	0.839	5.23 (1.48)
Decision Confidence (DC) C-Alpha = 0.921 AVE = 0.796 CR = 0.921 MSV = 0.705	- GenAI-generated insights enhance my confidence.	0.893	5.01 (1.59)
	- Based on GenAI insights, I focus on the real problem.	0.886	4.95 (1.60)
	- Based on GenAI insights, I feel confident about my decisions.	0.897	4.97 (1.95)
	- I found accurate information using GenAI for making academic decisions.	0.882	4.96 (1.52)
	- GenAI helps me to make quick decisions according to changes in the external environment.	0.912	4.95 (1.58)
Perceived Decision Capability (PC) C-Alpha = 0.917 AVE = 0.740 CR = 0.919 MSV = 0.705	- Updated GenAI information enhances my decision-making capability.	0.859	5.04 (1.51)
	- GenAI helps me to plan my activities.	0.783	4.95 (1.62)
	- GenAI seems dependable to me.	0.771	4.80 (1.59)
	- GenAI seems secure to me.	0.851	4.68 (1.59)
	- GenAI seems reliable to me.	0.895	4.96 (1.44)
GenAI Trust (GT) C-Alpha = 0.910 AVE = 0.722 CR = 0.912 MSV = 0.639	- I trust the GenAI outputs.	0.876	4.88 (1.47)

Note. Cronbach Alpha (C-Alpha), Average Variance Extract (AVE), Composite Reliability (CR), Maximum Shared Squared Variance (MSV), Items Mean & Standard Deviation (SD).

Table 2

Correlations among the constructs in the measurement model.

	GA	LR	IQ	DC	PC	GT	Mean	SD
GA	0.843						5.24	1.32
LR	0.665***	0.849					5.10	1.44
IQ	0.765***	0.633***	0.832				5.09	1.28
DC	0.608***	0.569***	0.675***	0.892			4.98	1.48
PC	0.642***	0.483***	0.695***	0.840***	0.860		4.97	1.39
GT	0.653***	0.466***	0.780***	0.730***	0.800***	0.849	4.83	1.35

Note. * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$; GA = GenAI Adoption; LR = Cognitive Load Reduction; IQ = Information Quality; DC = Decision Confidence; PC = Perceived decision capability; GT = GenAI Trust.

the AVEs and off-diagonal values represent the inter-construct correlations, the diagonal values were confirming the discriminant validity of the constructs in the model.

5.2. Structural model

The hypotheses and model fit of the study were tested using path analysis as shown in Table 3 and Fig. 2.

The results showed that H₁, H₂, H₃, and H₄ were all supported; that is, business students' adoption of GenAI significantly influenced their cognitive load reduction ($\beta = 0.76$, $t = 19.72$, $p < 0.01$), information quality ($\beta = 0.794$, $t = 26.33$, $p < 0.01$), decision confidence ($\beta = 0.706$, $t = 16.81$, $p < 0.01$), and perceived decision capability ($\beta = 0.194$, $t = 4.15$, $p < 0.01$). The study found an inverse relationship between cognitive load reduction and perceived decision capability of business students ($\beta = -0.181$, $t = -6.22$, $p < 0.01$), and thus H₅ was rejected. This implies that with the adoption of GenAI, students would become more dependent on AI, which would decrease their perceived capability to make academic or professional decisions. Lastly, information quality ($\beta = 0.166$, $t = 4.47$, $p < 0.01$) and decision confidence ($\beta = 0.722$, $t = 27.10$, $p < 0.01$) positively influenced business students' perceived decision capability, so both H₆ and H₇ were supported.

The model fit R² revealed that GenAI adoption explained 50.3% of the variance in cognitive load reduction, 64.3% in information quality, and 42.3% in decision confidence. Together, these four variables explained 82% of the variance in perceived decision capability.

5.2.1. Effect size

The individual effect size of the independent variables on the dependent variable, perceived decision capability, was assessed. As shown in Table 4, cognitive load reduction had a very high effect of 1.51, followed by a medium effect from GenAI adoption (between 0.15 and 0.30) and weak effects from information quality and decision confidence (< 0.15).

5.2.2. Mediation analysis

Process Macro Model was used to determine whether the relationship between GenAI adoption and decision-making capability was mediated by cognitive load reduction, information quality, or decision

Table 3

SEM hypotheses results.

Hypothesis	Path	β	S.E.	t-stat	Decision
H ₁	GA → LR	0.760	0.039	19.72***	Supported
H ₂	GA → IQ	0.794	0.030	26.33***	Supported
H ₃	GA → DC	0.706	0.042	16.81***	Supported
H ₄	GA → PC	0.194	0.047	4.15***	Supported
H ₅	LR → PC	-0.181	0.029	-6.22	Not Supported
H ₆	IQ → PC	0.166	0.037	4.47 ***	Supported
H ₇	DC → PC	0.722	0.027	27.10***	Supported

Note. * indicates $p < 0.05$, ** indicates $p < 0.01$, *** indicates $p < 0.001$, β = standardized regression weight, GA = GenAI Adoption; LR = Cognitive Load Reduction; IQ = Information Quality; DC = Decision Confidence; PC = Perceived decision capability; GT = GenAI Trust.

confidence. As shown in Table 5, all indirect relationships between GenAI adoption and perceived decision capability were mediated, except for cognitive load reduction ($\beta = 0.07$, LLCI = -0.03 , ULCI = 0.18).

5.2.3. Moderation analysis

As displayed in Table 6, the moderation effect of GenAI trust on the relationships between GenAI adoption and the three mediators (i.e., cognitive load reduction, information quality, and decision confidence) was assessed using Process Macro Model. First, the relationship between GenAI adoption and cognitive load reduction was significantly weakened by AI trust ($\beta = -0.08$, LLCI = -0.13 , ULCI = -0.04), thus supporting H_{8a}. That is, as trust in AI strengthens, the effect of GenAI adoption on cognitive load reduction substantially drops. This emphasises perceptions of GenAI trustworthiness, while acknowledging that these perceptions are not absolute but limited since GenAI retrieves and extracts publicly available information and then generates unique decisions. Second, GenAI trust moderated the relationship between GenAI adoption and information quality ($\beta = 0.04$, LLCI = -0.07 , ULCI = -0.02), thereby supporting H_{8b}. The negative effect indicated deterioration in the relationship. Third, GenAI trust did not moderate the relationship between GenAI adoption and decision confidence ($\beta = -0.03$, LLCI = -0.07 , ULCI = 0.1); accordingly, H_{8c} was not supported.

6. Discussion

Underpinned by DAT and CLT, this study initially addressed RQ1. It examined the association between business students' adoption of GenAI and their cognitive load reduction, information quality, decision confidence, and perceived decision capability by testing four hypotheses (H₁–H₄). First, the results indicate a positive association between business students' GenAI adoption and load reduction (H₁ supported). That is, the use of GenAI signifies a reduction in the cognitive load of business students. Consistent with CLT, students' adoption of GenAI reduces their intrinsic and extraneous load and improves their germane load. This result is supported by Patac and Patac (2025) and Jose et al. (2025), who noted that the use of GenAI (ChatGPT) in academia helps reduce students' cognitive load by assisting them in acquiring and applying knowledge to complete their complex tasks promptly and efficiently (Zhang et al., 2026). The study also found that overreliance on AI reduces students' critical thinking while raising concerns about data privacy and academic integrity (Vieriu & Petrea, 2025). This was verified by the current study, which found a negative relationship between load reduction and students' perceived decision capability.

Second, this study found a positive association between business students' GenAI adoption and information quality (H₂ supported). This suggests that GenAI enables students to extract high-quality information, thereby enhancing their cognitive abilities and improving their academic learning (Almugren, Rai, Chotia, et al., 2026). In line with CLT, this result also indicates that the use of advanced technologies (GenAI) can reduce working memory load among students, thereby strengthening their cognitive abilities and improving their learning efficiency (Paas & van Merriënboer, 2020). This finding is supported by Jose et al. (2025) and Almugren, Rai, Chotia, et al. (2026) who

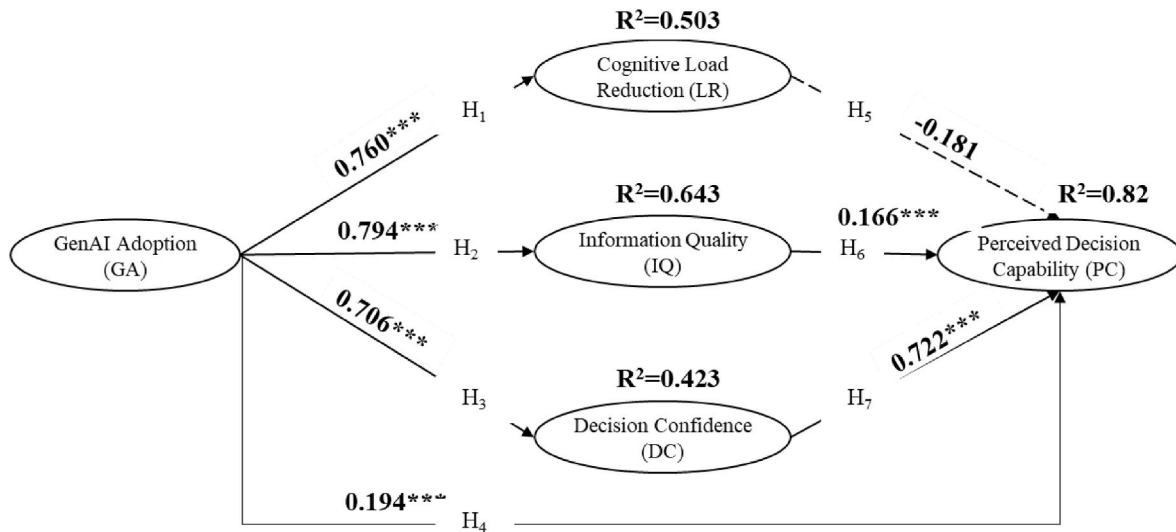


Fig. 2. Results of the structural model and hypotheses testing

Note. Solid arrow = significant relationship; * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

Table 4
Effect size.

Exogenous variable	Endogenous variable	R ²	R ² Excluded	Effect Size	Result
LR	PC	0.82	0.558	1.51	High
IQ	PC	0.82	0.823	0.01	Weak
DC	PC	0.82	0.801	0.13	Weak
GA	PC	0.82	0.788	0.20	Medium

Note. GA = GenAI Adoption; LR = Cognitive Load Reduction; IQ = Information Quality; DC = Decision Confidence; PC = Perceived decision capability.

Table 5
Mediation analysis (through process macro model).

Hypothesis	Path	β	S.E.	LLCI	ULCI	Result
H _{1a}	GA → LR → PC	0.07	0.05	-0.03	0.18	Not Supported
H _{2b}	GA → IQ → PC	0.42	0.06	0.30	0.53	Supported
H _{3c}	GA → DC → PC	0.51	0.05	0.41	0.61	Supported

Note: GA = GenAI Adoption; LR = Cognitive Load Reduction; IQ = Information Quality; DC = Decision Confidence; PC = Perceived decision capability.

concluded that GenAI meets students' informational needs and amplifies their cognitive abilities, thereby improving their learning engagement and helping them develop academic and professional skills ([Almugren et al., 2026](#)). However, some scholars have raised concerns about the accuracy, completeness, and bias of information generated through GenAI ([Chan & Hu, 2023](#); [Hannigan et al., 2024](#)), so it remains essential to train and educate students on the ethical and responsible use of GenAI in their academic endeavours through workshops and learning labs ([Jose et al., 2024](#)).

Third, hypothesis H₃ was validated, indicating a positive association between business students' adoption of GenAI and their decision

confidence. In conjunction with DAT, this result shows that GenAI augments decision confidence among business students. Previous studies have found similar evidence, indicating that the use of GenAI improves students' knowledge-building capacity ([Chen et al., 2023](#); [Zhang et al., 2026](#)), which in turn enhances their academic and professional learning efficiency and develops their confidence in decision-making. In the context of GenAI, this result is supported by [Al-Hattami's \(2025\)](#) and [Chan and Lee's \(2023\)](#) studies, which showed that across diverse situations, GenAI provides substantial value to students by delivering real-time information that enables them to make better decisions. Contrastingly, [Chong et al. \(2022\)](#) argued that if individuals lose their trust in GenAI, this could negatively influence their decision confidence. Additionally, [Oblakovic et al. \(2025\)](#) argued that the use of GenAI enhances users' metacognitive confidence while increasing their false confidence.

Fourth, the study found a positive association between business students' GenAI adoption and their perceived decision capability (H₄ supported). In accordance with CLT and DAT, the use of GenAI significantly reduces the mental effort (cognitive load) and working memory load required to process lengthy and complex academic information, thereby allowing students to learn and focus on strategic thinking ([Albashrawi, 2025](#)). [Al-Emran et al. \(2025\)](#) and [Li et al. \(2025\)](#) supported this association and concluded that individuals' confidence in GenAI enhances their self-confidence, thereby improving their perceived capability to make decisions based on AI-generated inputs. Similar support for the relationship was provided by [Phillips-Wren and Virvou \(2025\)](#); however, their study also raised concerns about data accuracy, privacy, manipulation, bias, and ethical use that influence the decision-making capabilities of its users.

Next, concerning RQ2, the study examined how cognitive load reduction, information quality, and decision confidence independently influence business students' perceived decision capability by testing three additional hypotheses (H₅₋₇). Cognitive load reduction is considered a prime variable in the hypothesised model because of its larger

Table 6
Moderating effect of GenAI Trust (Through Process Macro Model).

Hypothesis	Path	β	t-stat	p-value	LLCI	ULCI	Result
H _{8a}	GA and LR	-0.08	-3.64	0.00	-0.13	-0.04	Supported
H _{8b}	GA and IQ	-0.04	-3.05	0.00	-0.07	-0.02	Supported
H _{8c}	GA and DC	-0.03	-1.37	0.17	-0.07	0.01	Not Supported

Note. GA = GenAI Adoption; LR = Cognitive Load Reduction; IQ = Information Quality; DC = Decision Confidence.

effect size (1.51) on perceived decision capability. The results reject H_5 and quantify a negative association between cognitive load reduction and business students' perceived decision capability, suggesting that adopting GenAI diminishes business students' critical thinking and, consequently, their perceived decision capability. That is, while GenAI may reduce extraneous load and improves germane load, overreliance on GenAI diminishes students' critical thinking and hampers their perceived decision capability. This extends DAT to the context of students' adoption of GenAI in academia. When students use GenAI tools as substitutes rather than amplifiers to augment their cognition, they rely passively on AI outputs, which weakens their perceived decision capability. This result is supported by [Bai et al. \(2023\)](#), who ascertained that the overuse of GenAI undermines students' critical thinking ([Jose et al., 2025](#)). This study extends this by showing that the use of GenAI reduces students' cognitive load; however, excessive use impairs their perceived decision capability, a unique contribution to AI literature.

Regarding H_6 , the results indicate that information quality has a significant positive influence on business students' perceived decision capability. That is, the qualitative information generated through GenAI enhances students' capability to analyse complex scenarios, appraise academic alternatives effectively, and make rational decisions ([Almogren et al., 2024](#)). This finding aligns with DAT, which asserts that reliable and context-specific information supports cognitive functions that augment students' perceived decision capability ([Albashrawi, 2025](#); [Almogren et al., 2024](#)). Preceding researchers have emphasised the benefits of GenAI in terms of the quality of the information it generates, but they have also raised concerns about information bias, ethical use, and accuracy that might affect students' perceived ability to make decisions ([Phillips-Wren & Virvou, 2025](#); [Vieriu & Petrea, 2025](#)).

In testing the last hypothesis (H_7), the results reveal a positive influence of decision confidence on business students' perceived decision capability. This indicates that as students become more confident in their academic decisions, their perceived decision-making efficiency increases. [Li et al. \(2025\)](#) supported this finding by showing that people's confidence in GenAI enhances their self-confidence, which, in turn, affects their decision capability. This result contradicts the findings of [Oblakovic et al. \(2025\)](#), who reported that the use of GenAI strengthens students' metacognitive confidence while simultaneously diminishing their decision-making capability. Chong et al. (2022) stated that individuals' confidence in themselves is the key factor (not their confidence in GenAI) that encourages them to accept GenAI's suggestions and that impacts their decisions.

The mediation hypotheses (H_{1a} , H_{2b} , and H_{3c}) were formulated to answer RQ3. Unlike cognitive load reduction, both information quality and decision confidence are acknowledged to be partial mediators, as evidenced by their influence on the relationship between business students' GenAI adoption and perceived decision capability. The quality of information generated through GenAI has been shown to promote thoughtfulness in students' decision-making processes. This result is supported by [Changalima et al. \(2024\)](#), who affirmed that information quality generated through GenAI (ChatGPT) strengthens students' behavioural intentions. Furthermore, [Albashrawi \(2025\)](#) asserted that GenAI-based information improves decision confidence among business students, which significantly enhances their perceived academic or entrepreneurial decision capabilities ([Almogren, Rai, Chotia, et al., 2026](#); [Almogren et al. 2026a](#)). According to [Gkintoni et al. \(2025\)](#), the use of GenAI can improve the learning efficiency of students and reduce their cognitive load; however, concerns about data security, ethical considerations, and trustworthiness may affect their decision-making capability. In summary, the results of the mediation analysis show that GenAI can reduce students' cognitive load but this does not influence their perceived decision capability.

Finally, the study investigated the moderating role of GenAI trust in the hypothesised relationship between GenAI adoption and three key outcomes on business students: cognitive load reduction, information quality, and decision confidence. The findings reveal that GenAI trust

moderates cognitive load reduction (H_{8a} supported) and quality information (H_{8b} supported) but not decision confidence (H_{8c} not supported). That is, as students' trust in GenAI increases, the impact of GenAI adoption on reducing cognitive load becomes stronger, and perceptions of information quality are improved. This first impact can be rephrased as follows: Students who trust GenAI are more likely to experience reduced mental effort when using GenAI tools. This moderation analysis can be regarded as an extension of the research by [Patac and Patac \(2025\)](#) and [Jose et al. \(2025\)](#) due to the role of GenAI trust in students' GenAI adoption and cognitive load reduction. The second impact can also be observed among students with greater trust in GenAI, who are more likely to perceive the information generated by GenAI as accurate, reliable, relevant, and useful ([Lyu et al., 2025](#)). Conversely, when GenAI trust is low, even with high GenAI adoption, students may remain sceptical about the quality of GenAI-generated information ([Kim et al., 2023](#)). This relationship emphasises the critical role of trust in shaping users' perceptions of GenAI effectiveness, especially in academic or decision-making contexts. Furthermore, simply adopting any GenAI tool is insufficient to achieve better information quality. Users of these tools must also trust the GenAI's credibility to fully benefit from the quality of the information they provide. Therefore, building trust in GenAI tools through transparency, accuracy, and user experience is essential for maximising the impact of GenAI in education. Lastly, this study found that GenAI trust does not moderate the relationship between GenAI adoption and decision confidence. This result suggests that students' level of trust in GenAI has no significant influence on how GenAI affects their confidence in decision-making. One possible interpretation is that decision confidence may stem more directly from the perceived usefulness or clarity of the output rather than from generalised trust in the technology itself. Even if students trust GenAI, if the outputs are complex, vague, or contextually misaligned, their confidence in making a decision may remain unaffected ([Chan & Zhou, 2023](#)).

7. Implications

7.1. Theoretical implications

The study offers potential theoretical implications when viewed through the lenses of DAT and CLT. Based on DAT, the adoption of GenAI can augment business students' perceived decision capability by facilitating their access to relevant, timely, and high-quality information ([Tao et al., 2024](#)). This is reinforced by the strong connection between students' GenAI adoption and decision confidence, information quality, and perceived decision capability. This study extends DAT by examining decision confidence and information quality as leading indicators of students' adoption of GenAI for improving their perceived decision capability. GenAI serves as a catalyst, enhancing students' confidence and information perspicuity so that they can make decisions swiftly and effectively ([Leyer & Schneider, 2021](#)). It also expands their cognitive boundaries and fortifies their perceived decision-making ([George & Wooden, 2023](#)). For instance, if teaching methods such as case studies and real-time simulations are integrated with GenAI tools, business students' intellectual abilities may be enhanced, and more conclusive decisions on their academic assignments, learning development, and professional outlook may be promoted ([Nguyen & Elbanna, 2025](#)).

CLT posits that reducing cognitive load enables users (students) to process information more efficiently ([De Jong, 2010](#)). This study found that GenAI adoption significantly reduces business students' cognitive load, yet this reduction negatively affects their perceived decision capability. This indicates that GenAI reduces mental strain, but overreliance may diminish active cognitive engagement, thereby undermining decision-making competence—a paradox explained by CLT's concern with underload or disengagement ([Bai et al., 2023](#); [Cela et al., 2024](#); [Jose et al., 2025](#)).

The mediation analysis reveals that GenAI's influence on perceived

decision capability is significantly mediated by information quality and decision confidence, but not by cognitive load reduction. This suggests that GenAI supports decision-making primarily through knowledge augmentation rather than stress relief. Similarly, the moderation analysis reveals that GenAI trust weakens the relationship between GenAI adoption and both cognitive load reduction and information quality, underscoring that overreliance on GenAI may reduce students' critical thinking skills (Cela et al., 2024).

7.2. Practical implications

This study offers several practical implications for students, educators, educational institution officials, and industry stakeholders. For business students, adopting GenAI could improve information quality and decision confidence, further enhancing their perceived decision capability (Albashrawi, 2025). This suggests that integrating GenAI tools into educational programmes and training workshops could significantly enhance students' ability to make well-informed decisions. By exposing students to AI-driven simulations, scenario analyses, or data interpretation tools, educational institutions can help students cultivate greater confidence and sharper analytical skills. However, students must learn to use GenAI as a support system (Jose et al., 2025). The negative association between load reduction and perceived decision capability indicates that over-reliance on GenAI may lead to passive learning or reduced critical thinking (Vieriu & Petrea, 2025). Therefore, students should be trained to remain cognitively engaged with GenAI tools. They should become competent at validating GenAI-based outputs, and that competence should be reflected in their decisions.

The study's results suggest that a radical shift in educators' teaching practices is needed. GenAI should be integrated as a learning companion to promote decision augmentation rather than substitution (Cela et al., 2024). Educators can use GenAI-driven tools like chatbots, content summarisers, and data analysers to create decision-making scenarios in which students analyse and justify GenAI recommendations. Educators should train students not only to use GenAI tools but also to interpret, validate, and critically assess AI-generated outputs, thereby building a stronger foundation of trust and confidence in their decision-making (Li et al., 2025; Lyu et al., 2025). Since decision confidence plays a vital role in this, educators should encourage students to develop trust in GenAI through practice. Furthermore, educators should emphasise GenAI's limitations to prevent blind dependence (Bai et al., 2023; Chong et al., 2022; Phillips-Wren & Virvou, 2025).

Educational institution officials could redesign curriculum frameworks to incorporate GenAI literacy, ethical reasoning, and critical evaluations of AI-generated outputs (Fischer et al., 2024). Educational institutions should also invest in infrastructure that enables the safe, interactive use of GenAI tools across disciplines (Gupta, Mahajan, et al., 2024). Additionally, the finding of this study suggest the need for a balanced cognitive load. Therefore, while GenAI should be used to simplify complex tasks, course design should be modified to encourage students to think independently. Moreover, officials should facilitate initiatives such as awareness programmes, research labs, and GenAI-related workshops to help students and educators learn to use GenAI ethically in an academic context (Wood & Moss, 2024).

In terms of practical implications relevant to industry, especially for organisations that hire business graduates at a large scale, the findings emphasise the importance of GenAI-enabled productivity platforms led by well-trained professionals. Accordingly, organisations need to work with academic institutions to deliver GenAI-based learning workshops that animate decision-making situations in a work environment. Qualified industry professionals can also deliver training modules for business students, teaching them to use GenAI in real-time scenarios while maintaining cognitive engagement and accountability. The focus, however, should be on teaching professionals how to use GenAI tools to obtain precise and reliable information because decision confidence and information quality mediate GenAI's outcomes.

8. Conclusions

This study drew on concepts from DAT and CLT to emphasise the significant role of GenAI in enhancing business students' perceptions of their decision-making capability. The findings show that GenAI positively influences business students' cognitive load reduction, information quality, and decision confidence. Consequently, these three factors are considered central to improving business students' perceived decision-making competence. Next, the study found decision confidence and information quality to be potent mediators. Hence, GenAI helps business students make better decisions by improving the clarity and reliability of the information they use. However, the negative impact of cognitive load reduction on perceived decision capability suggests the potential risk of overreliance on GenAI, which may lead to cognitive disengagement (Bai et al., 2023; Cela et al., 2024). Additionally, the moderating role of GenAI trust reveals that excessive trust can reduce GenAI's effectiveness, especially in improving information quality or managing cognitive load (Lyu et al., 2025). Overall, while GenAI is a valuable support tool for students, its benefits depend on a balance between reasonable usage, the use of critical thinking skills, and responsible trust.

This study had some limitations that warrant attention. First, it adopted a quantitative approach; however, this may have prevented the study from fully capturing the depth of students' attitudes or cognitive behaviours related to GenAI adoption. Qualitative or mixed-methods research could provide a clearer understanding of students' use of GenAI and its impact on their decisions. Second, the study did not incorporate non-AI control conditions or direct measures of academic integrity, which may have limited claims about comparative performance with and without GenAI. Future work should include experimental comparisons (AI vs. no-AI conditions), pre- and post-learning measures, and direct measures of academic integrity to empirically test these implications. Furthermore, while this study captured students' perceived decision capability, future research could integrate objective performance indicators such as task accuracy, academic outcomes, or experimental decision-making tasks to examine the alignment between perceived and actual decision quality when using GenAI tools. Third, the research sample was confined to a specific educational category within certain areas, namely business university students in the United Arab Emirates, limiting generalisability to other educational institutions, or geographical areas. Further research could explore other discipline-specific applications of GenAI (e.g., healthcare, engineering, or humanities) across countries to assess whether cognitive and decision-making impacts among students vary. Lastly, the study identified GenAI trust as the only moderator variable. In future work, other factors, such as ethical concerns, technological anxiety, and digital literacy, could be considered to examine their influence on student judgement.

CRedit authorship contribution statement

Jaskirat Singh Rai: Writing – review & editing, Writing – original draft, Supervision, Project administration, Methodology, Formal analysis, Conceptualization. **Sakshi Kathuria:** Writing – review & editing, Writing – original draft, Resources, Methodology, Formal analysis. **Harpominder Kaur:** Writing – review & editing, Writing – original draft, Resources, Methodology, Formal analysis. **Amanpreet Singh:** Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Maher N. Itani:** Writing – review & editing, Writing – original draft, Visualization, Resources, Methodology, Conceptualization.

Ethics declaration

This study was conducted in accordance with the guidelines of the Research Ethics Committee of Ajman University (approval code B-F-H-

13-Oct). Participants provided consent to participate in this study via Google Forms. All data was kept confidential and accessible only by the researchers.

Availability of data and materials

The datasets used and/or analysed during the current study are available from the corresponding author on reasonable request.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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